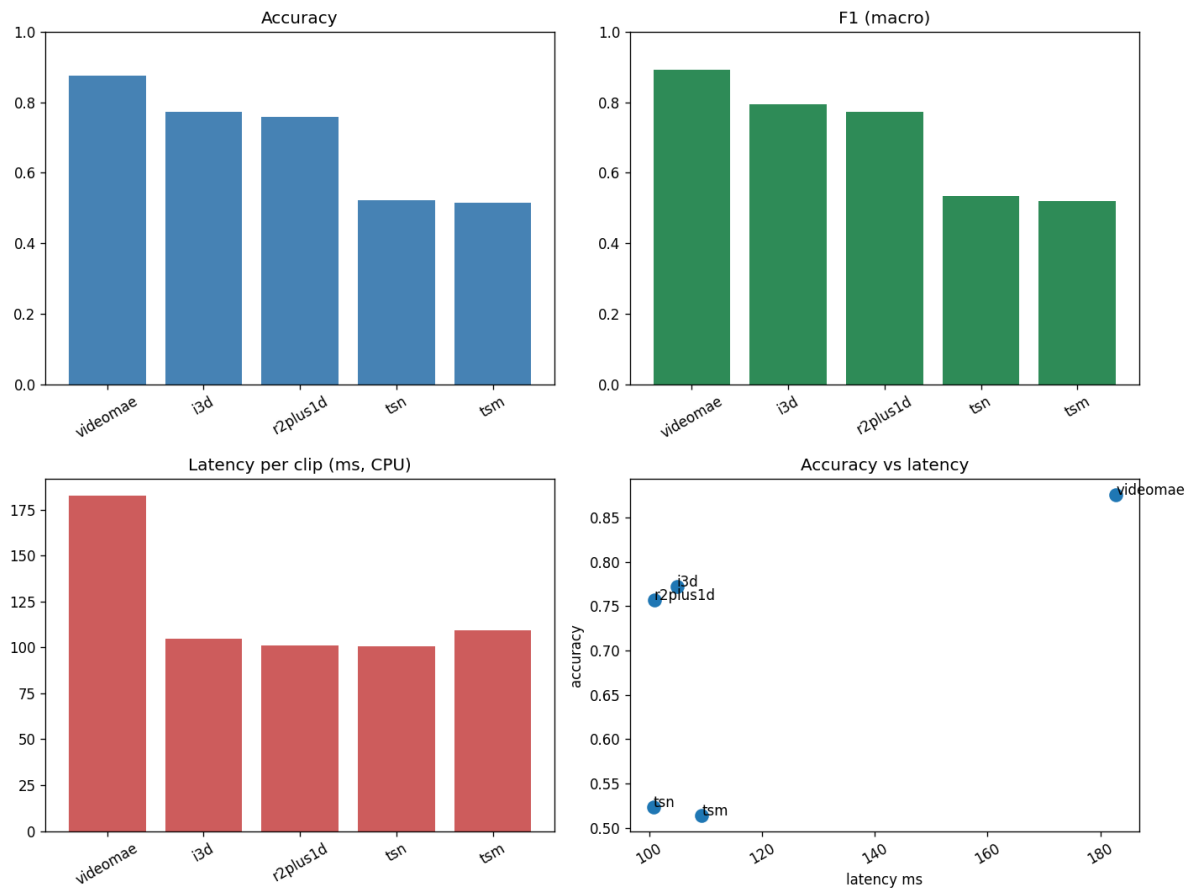


Action Recognition — Model Comparison Report

Dataset: HMDB51 subset (10 classes). Classes: climb, draw_sword, fencing, golf, pullup, ride_bike, run, shoot_bow, walk, wave.

model	accuracy	precision_macro	recall_macro	f1_macro	latency_s_per_clip	fps	stability	mean_confidence	params	size_mb
videomae	0.876	0.894	0.892	0.892	0.183	87.543	1.0	0.664	86234890	328.96
i3d	0.772	0.825	0.781	0.795	0.105	152.484	0.95	0.476	27244362	104.132
r2plus1d	0.757	0.79	0.772	0.772	0.101	158.603	0.85	0.443	31305255	119.517
tsn	0.523	0.552	0.536	0.535	0.101	79.436	0.9	0.295	11181642	42.691
tsm	0.514	0.554	0.52	0.521	0.109	73.203	0.862	0.265	11181642	42.691

Comparison charts



Confusion matrix — tsn

Confusion matrix — tsn

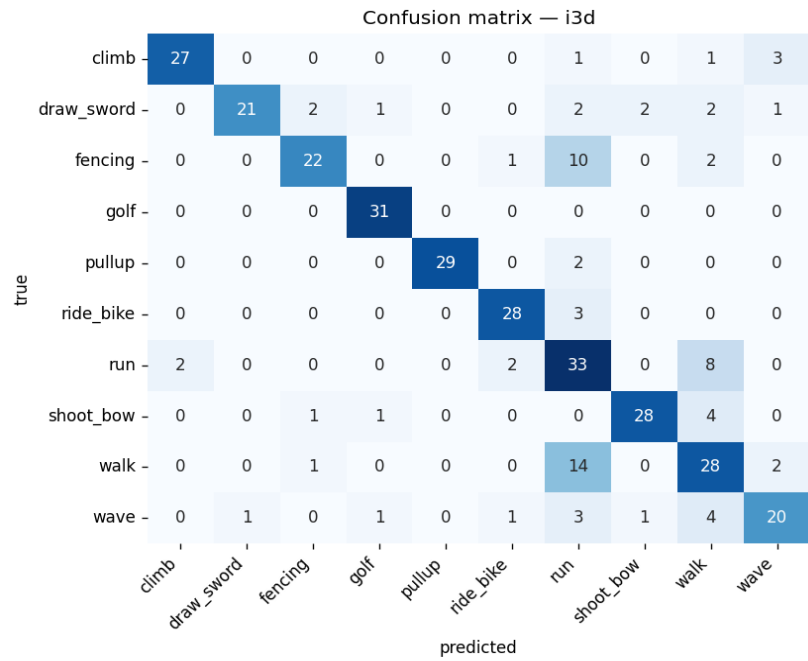
true	climb	26	0	0	1	1	0	1	1	2	0
	draw_sword	1	14	4	1	3	0	1	0	3	4
	fencing	3	2	11	1	0	2	6	0	7	3
	golf	0	0	0	29	1	0	1	0	0	0
	pullup	1	0	1	1	18	0	4	2	3	1
	ride_bike	1	0	2	1	2	19	1	3	2	0
	run	0	2	2	3	1	1	12	2	20	2
	shoot_bow	1	0	0	1	3	0	3	23	3	0
	walk	1	1	1	1	0	1	11	0	23	6
	wave	1	2	4	1	5	0	4	1	7	6
		climb	draw_sword	fencing	golf	pullup	ride_bike	run	shoot_bow	walk	wave
		predicted									

Confusion matrix — tsm

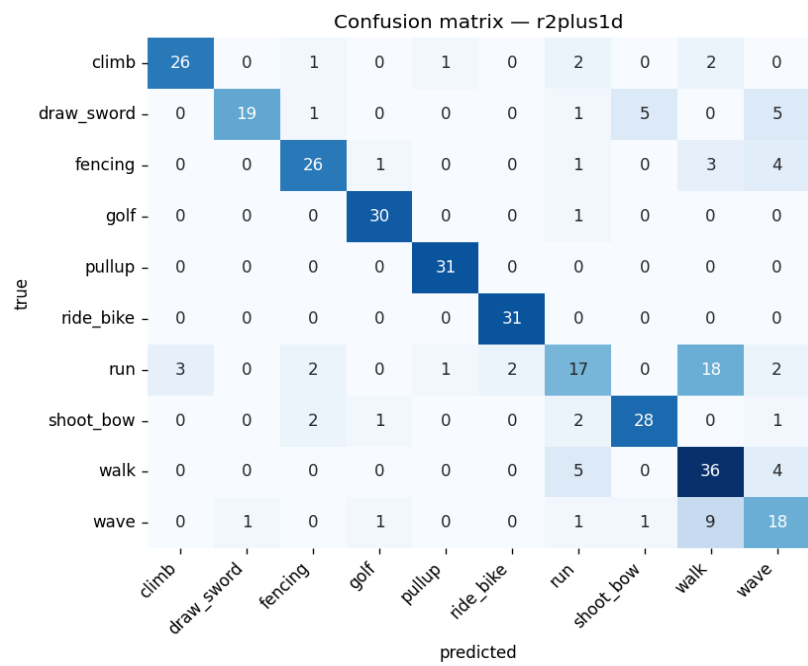
Confusion matrix — tsm

true	climb	24	0	0	1	0	1	1	0	5	0
	draw_sword	0	12	2	2	3	1	1	1	7	2
	fencing	2	1	9	1	2	1	4	0	13	2
	golf	0	0	0	29	2	0	0	0	0	0
	pullup	2	1	0	0	18	0	3	3	4	0
	ride_bike	3	0	4	4	1	15	1	1	2	0
	run	2	0	2	0	1	3	18	1	16	2
	shoot_bow	1	1	0	2	2	0	3	22	2	1
	walk	0	2	1	1	1	1	13	0	24	2
	wave	0	1	3	1	6	1	3	1	8	7
		climb	draw_sword	fencing	golf	pullup	ride_bike	run	shoot_bow	walk	wave
		predicted									

Confusion matrix — i3d



Confusion matrix — r2plus1d



Confusion matrix — videomae

Confusion matrix — videomae

	climb	draw_sword	fencing	golf	pullup	ride_bike	run	shoot_bow	walk	wave
climb	31	0	0	0	0	0	1	0	0	0
draw_sword	0	29	1	0	0	0	0	0	0	1
fencing	0	0	33	0	0	0	1	0	1	0
golf	0	0	0	31	0	0	0	0	0	0
pullup	0	0	0	0	31	0	0	0	0	0
ride_bike	0	0	0	0	0	30	0	0	1	0
run	0	0	3	0	0	0	33	0	8	1
shoot_bow	0	0	0	1	0	0	0	31	1	1
walk	0	0	0	0	0	0	12	0	28	5
wave	1	0	1	0	0	0	0	0	3	26